SALES DATA ANALYSIS

PREPARED BY

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TOOLS AND TECHNOLOGY USED

EXCEL /POWERQUERY/POWERBI

2026

1. Project Introduction

The Sales Data Analysis project focuses on analyzing sales performance using Microsoft Excel and Microsoft Power BI.

The main purpose of this project is to transform raw sales data into meaningful business insights through data cleaning, data transformation, data analysis, and interactive data visualization.

Microsoft Excel was used for data cleaning, calculations, Pivot Tables, Pivot Charts, and dashboard creation.

Microsoft Power BI was used for advanced data transformation, data modelling, DAX calculations, KPI development, and interactive dashboard creation.

This project helps understand sales trends, regional performance, category-wise sales, customer behaviour, discount impact, and profitability.

The final dashboards provide a clear and interactive view of business performance to support data-driven decision-making.

2. Project Objectives

The main objectives of this Sales Data Analysis project are:

1. To analyze overall sales performance and business profitability.
2. To clean and organize raw sales data using Microsoft Excel and Power Query.
3. To identify monthly and yearly sales trends.
4. To analyze sales performance across different regions and cities.
5. To identify top-performing product categories and sub-categories.
6. To evaluate the impact of discounts on sales and profitability.
7. To analyze customer segments and sales channels.
8. To calculate important business KPIs using Excel formulas and Power BI DAX measures.
9. To create interactive dashboards using Excel and Power BI.
10. To generate meaningful business insights that support better decision-making.

3. Dataset Description

The dataset used in this project contains sales transaction details that help analyze business performance, customer information, product sales, revenue, discounts, and profitability.

Each record represents a sales transaction containing information about orders, customers, products, locations, sales amounts, and customer satisfaction.

The dataset was used for analysis in both Microsoft Excel and Microsoft Power BI.

3.1 Dataset Columns

The dataset contains the following columns:

1. Order ID
2. Order Date
3. Customer ID
4. Region
5. City
6. Segment
7. Channel
8. Category
9. Sub-Category
10. Product ID
11. Quantity
12. Unit Price
13. Discount
14. Retail Sales
15. Discount Amount
16. Net Sales
17. Cost
18. Profit
19. Shipping Days
20. Return
21. Customer Satisfaction
22. Payment Method
23. Profit Margin
24. Average Selling Price

3.2 Dataset Overview

The dataset includes different types of information:

- Order Information: Order ID and Order Date.
- Customer Information: Customer ID and Segment.
- Product Information: Category, Sub-Category and Product ID.
- Geographical Information: Region and City.
- Sales Information: Quantity, Unit Price, Retail Sales and Net Sales.
- Financial Information: Discount, Discount Amount, Cost, Profit and Profit Margin.
- Customer Information: Return, Customer Satisfaction and Payment Method.
- Shipping Information: Shipping Days.

3.3 Dataset Summary

- The sales dataset consists of 1,000 records and 24 columns.

Dataset Attribute	Description
Project Name	Sales Data Analysis
Total Records	1,000
Total Columns	24
File Format	Excel (.xlsx)
Data Type	Structured Sales Data
Primary Tools	Microsoft Excel and Power BI
Analysis Type	Sales and Profitability Analysis

- The dataset provides sufficient information to perform sales analysis, customer analysis, regional analysis, product performance analysis, discount analysis, and profitability analysis.

3.4TOOLS AND TECHNOLOGY USED

Microsoft Excel: Data cleaning, Formulas, Pivot Tables, Pivot Charts, Dashboard

Power Query: Data Transformation, Advanced Data Cleaning

PowerBI: Data Modelling, Dax Measures, KPI Cards, Interactive Dashboard

4. Tools and Technologies Used

The Sales Data Analysis project was developed using Microsoft Excel and Microsoft Power BI. These tools were used for data cleaning, data transformation, analysis, visualization, and dashboard development.

4.1 Microsoft Excel

Microsoft Excel was used for initial data preparation and sales analysis.

Features used:

- Data Cleaning and Formatting
- Excel Formulas and Functions
- Sorting and Filtering
- Pivot Tables
- Pivot Charts
- KPI Analysis
- Interactive Slicers
- Sales Dashboard Creation

4.2 Microsoft Power Query

Power Query was used to clean and transform the sales dataset before analysis.

Features used:

- Removing Errors
- Handling Missing Values
- Changing Data Types
- Data Transformation
- Column Formatting
- Data Preparation

4.3 Microsoft Power BI

Microsoft Power BI was used to develop interactive dashboards and perform advanced sales analysis.

Features used:

- Power Query Editor
- Data Modelling
- Relationships
- DAX Measures
- KPI Cards
- Bar Charts

- Donut Charts
- Scatter Charts
- Decomposition Tree
- Slicers and Filters
- Interactive Dashboard Development

4.4 DAX (Data Analysis Expressions)

DAX was used to create calculated measures for business performance analysis.

Examples:

- Total Sales
- Total Profit
- Total Quantity
- Total Orders
- Average Selling Price
- Profit Margin
- Discount Analysis

4.5 Microsoft Word

Microsoft Word was used to prepare the project documentation, including the project introduction, objectives, dataset description, methodology, analysis, screenshots, findings, and conclusion.

5. Data Cleaning and Preparation in Microsoft Excel

Microsoft Excel was used for the initial inspection, cleaning, and preparation of the sales dataset.

The main purpose of data cleaning was to ensure that the dataset was accurate, consistent, and ready for further analysis.

5.1 Data Inspection

- Opened the raw sales dataset in Microsoft Excel.
- Reviewed all 24 columns and 1,000 records.
- Checked column names and data formats.
- Identified potential data quality issues.

5.2 Duplicate Data Checking

- Checked the dataset for duplicate records.
- Used Excel's Remove Duplicates feature where required.
- Ensured that duplicate transactions did not affect the analysis.

5.3 Missing Value Checking

- Checked the dataset for blank and missing values.
- Reviewed missing values in important columns.
- Handled missing values appropriately wherever required.

5.4 Data Formatting

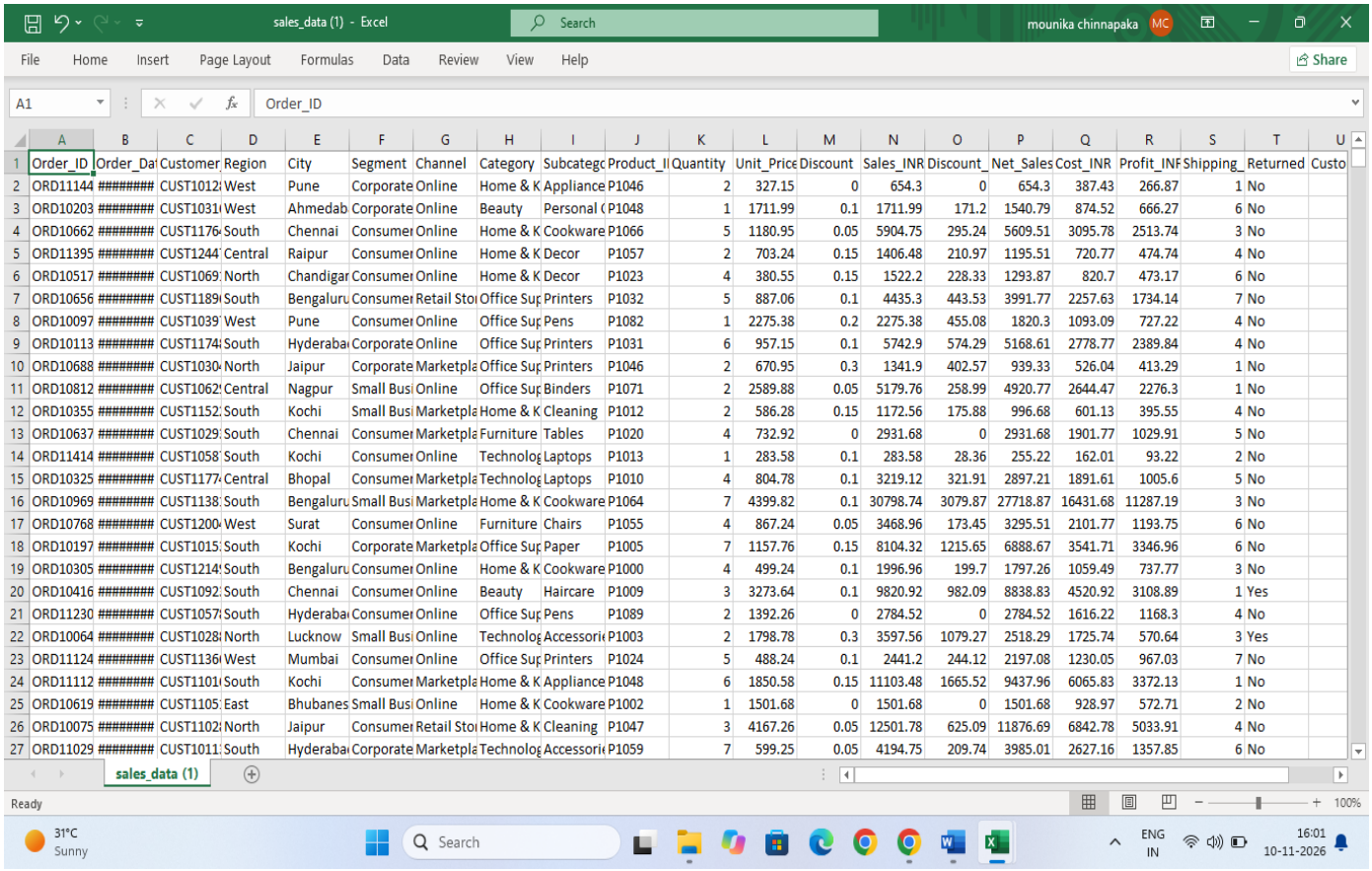
- Applied appropriate date formats to the Order Date column.
- Checked numerical data types for Quantity, Unit Price, Sales, Cost, and Profit.
- Ensured consistent formatting across the dataset.
- Applied appropriate number and currency formats.

5.5 Data Validation

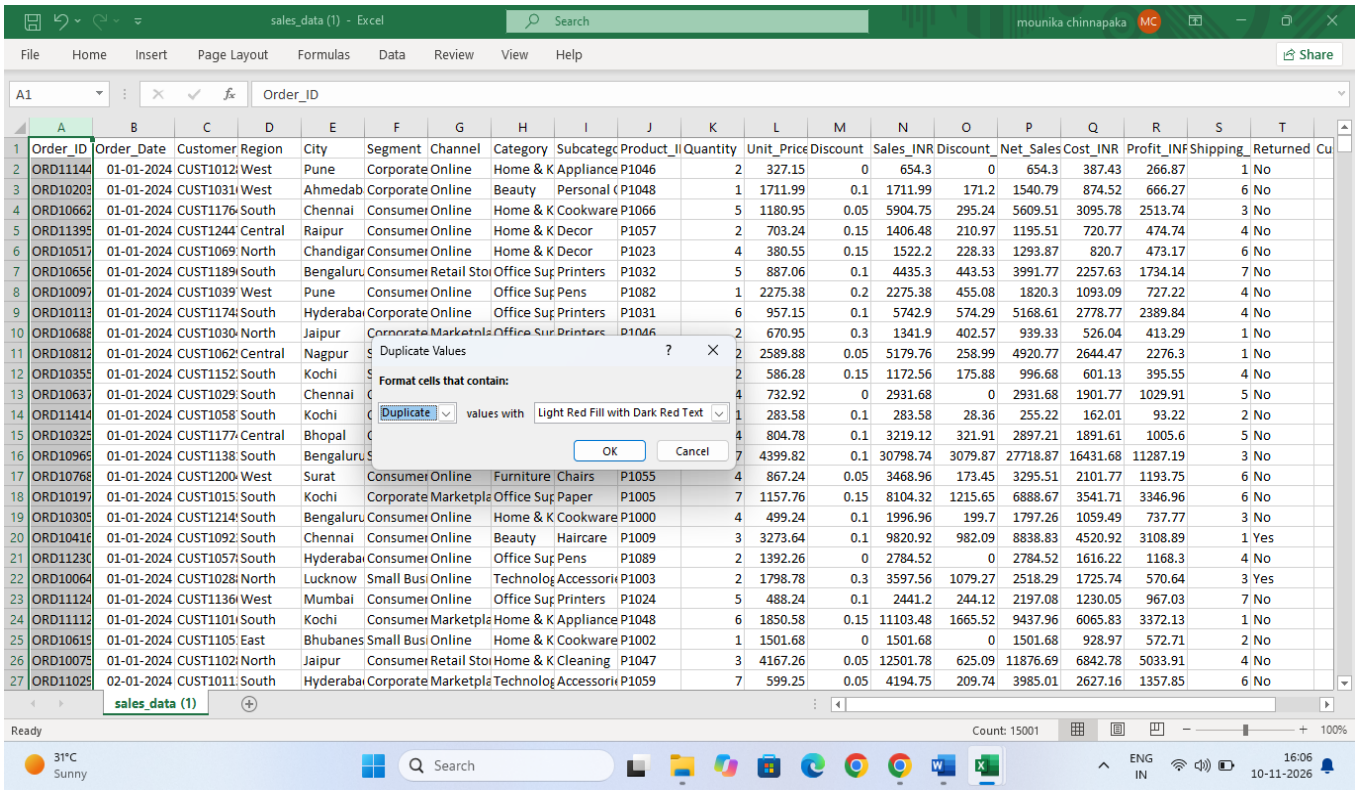
- Reviewed important columns for inconsistent or incorrect values.
- Checked sales and profit-related fields for accuracy.
- Verified that the dataset was suitable for Pivot Table analysis.

5.6 Final Dataset Preparation

After the initial cleaning and validation process, the dataset was prepared for further analysis using Excel Pivot Tables, Pivot Charts, and dashboard development.

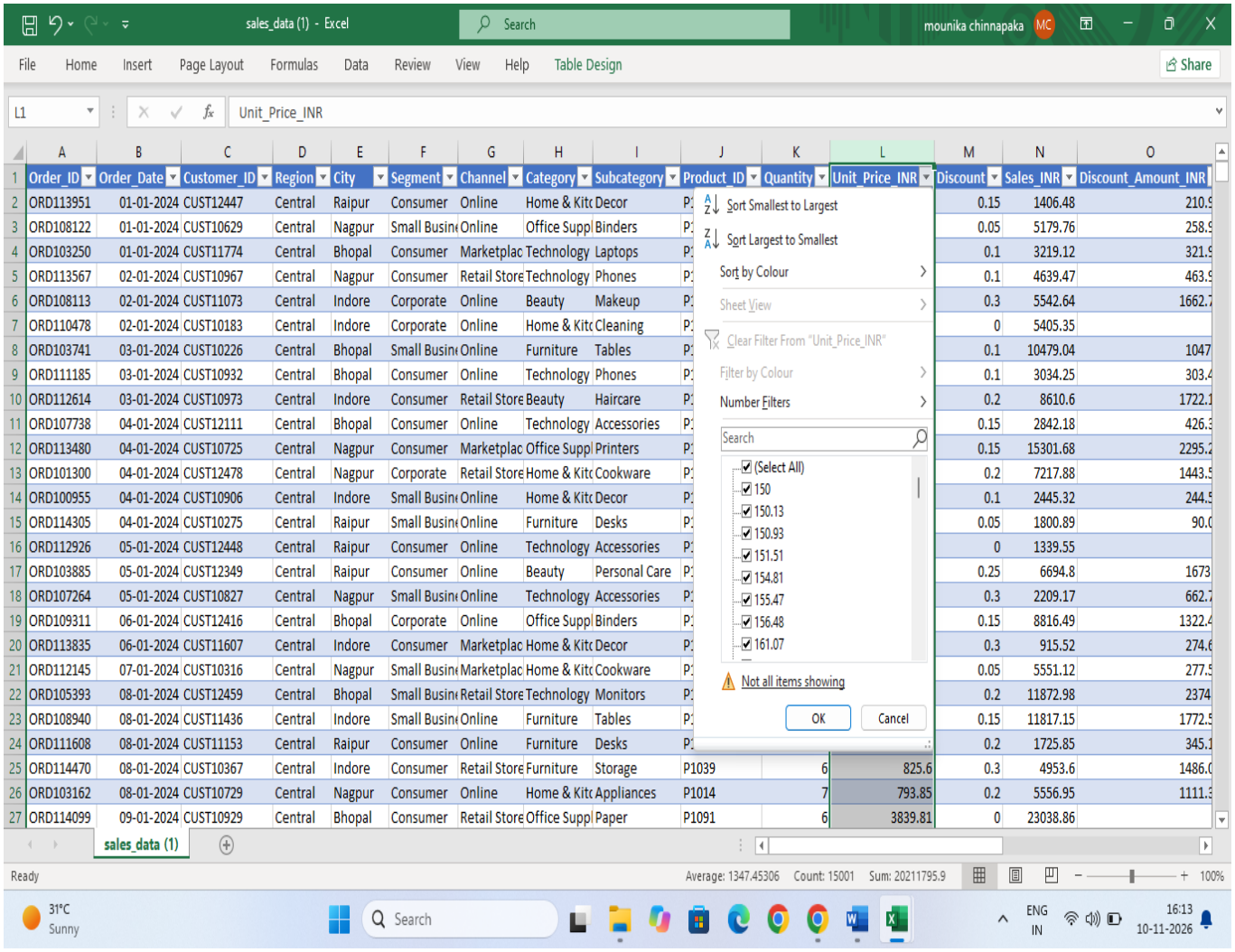


1.Sales Raw Data



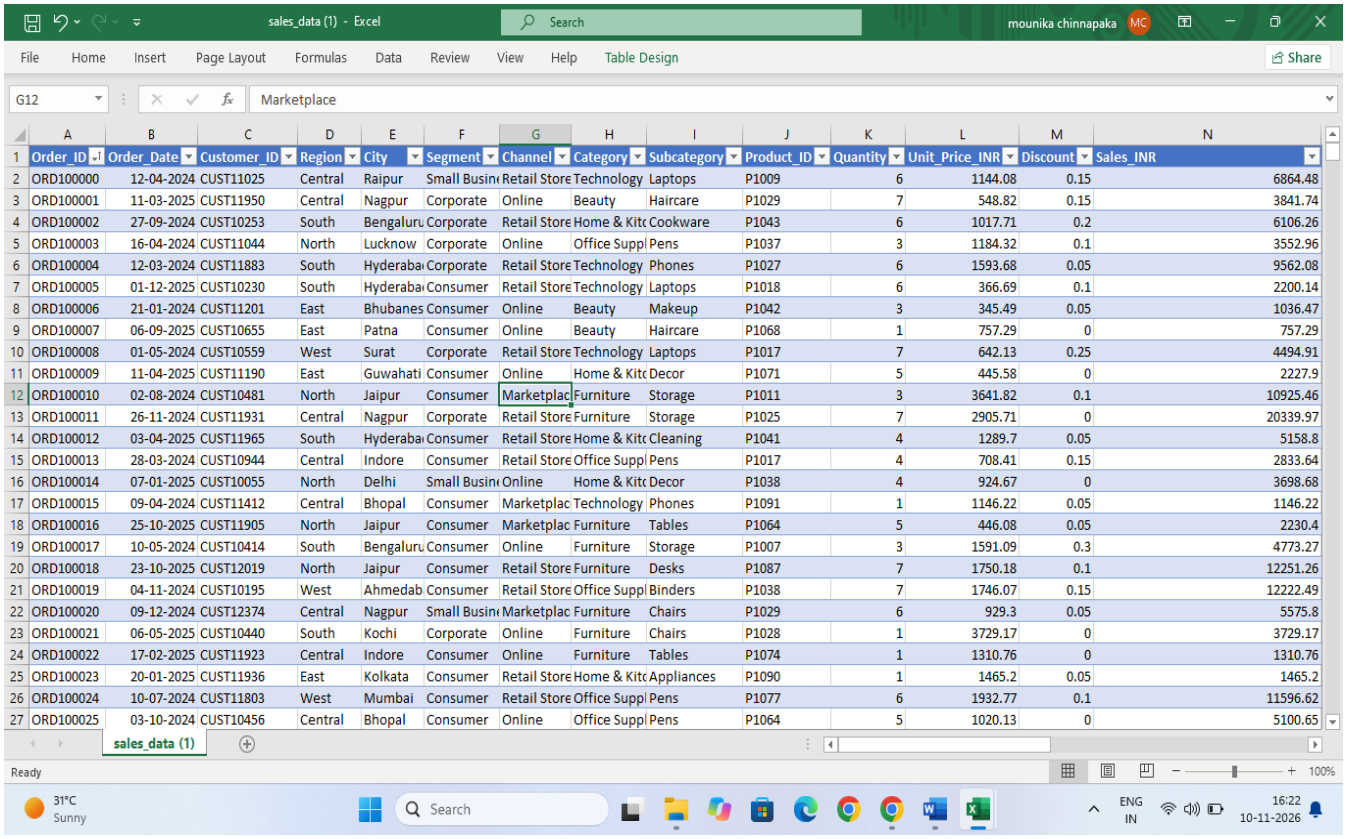
2.Duplicates checking

3.Missing Values Checking



Order_ID	Order_Date	Customer_ID	Region	City	Segment	Channel	Category	Subcategory	Product_ID	Quantity	Unit_Price_INR	Discount	Sales_INR	Discount_Amount_INR
ORD113951	01-01-2024	CUST12447	Central	Raipur	Consumer	Online	Home & Kitc	Decor	P:			0.15	1406.48	210.9
ORD108122	01-01-2024	CUST10629	Central	Nagpur	Small Busine	Online	Office Suppl	Binders	P:			0.05	5179.76	258.9
ORD103250	01-01-2024	CUST11774	Central	Bhopal	Consumer	Marketplac	Technology	Laptops	P:			0.1	3219.12	321.9
ORD113567	02-01-2024	CUST10967	Central	Nagpur	Consumer	Retail Store	Technology	Phones	P:			0.1	4639.47	463.9
ORD108113	02-01-2024	CUST11073	Central	Indore	Corporate	Online	Beauty	Makeup	P:			0.3	5542.64	1662.7
ORD110478	02-01-2024	CUST10183	Central	Indore	Corporate	Online	Home & Kitc	Cleaning	P:			0	5405.35	
ORD103741	03-01-2024	CUST10226	Central	Bhopal	Small Busine	Online	Furniture	Tables	P:			0.1	10479.04	1047
ORD111185	03-01-2024	CUST10932	Central	Bhopal	Consumer	Online	Technology	Phones	P:			0.1	3034.25	303.4
ORD112614	03-01-2024	CUST10973	Central	Indore	Consumer	Retail Store	Beauty	Haircare	P:			0.2	8610.6	1722.1
ORD107738	04-01-2024	CUST12111	Central	Bhopal	Consumer	Online	Technology	Accessories	P:			0.15	2842.18	426.3
ORD113480	04-01-2024	CUST10725	Central	Nagpur	Consumer	Marketplac	Office Suppl	Printers	P:			0.15	15301.68	2295.2
ORD101300	04-01-2024	CUST12478	Central	Nagpur	Corporate	Retail Store	Home & Kitc	Cookware	P:			0.2	7217.88	1443.5
ORD100955	04-01-2024	CUST10906	Central	Indore	Small Busine	Online	Home & Kitc	Decor	P:			0.1	2445.32	244.5
ORD114305	04-01-2024	CUST10275	Central	Raipur	Small Busine	Online	Furniture	Desks	P:			0.05	1800.89	90.0
ORD112926	05-01-2024	CUST12448	Central	Raipur	Consumer	Online	Technology	Accessories	P:			0	1339.55	
ORD103885	05-01-2024	CUST12349	Central	Raipur	Consumer	Online	Beauty	Personal Care	P:			0.25	6694.8	1673
ORD107264	05-01-2024	CUST10827	Central	Nagpur	Small Busine	Online	Technology	Accessories	P:			0.3	2209.17	662.7
ORD109311	06-01-2024	CUST12416	Central	Bhopal	Corporate	Online	Office Suppl	Binders	P:			0.15	8816.49	1322.4
ORD113835	06-01-2024	CUST11607	Central	Indore	Consumer	Marketplac	Home & Kitc	Decor	P:			0.3	915.52	274.6
ORD112145	07-01-2024	CUST10316	Central	Nagpur	Small Busine	Marketplac	Home & Kitc	Cookware	P:			0.05	5551.12	277.5
ORD105393	08-01-2024	CUST12459	Central	Bhopal	Small Busine	Retail Store	Technology	Monitors	P:			0.2	11872.98	2374
ORD108940	08-01-2024	CUST11436	Central	Indore	Small Busine	Online	Furniture	Tables	P:			0.15	11817.15	1772.5
ORD111608	08-01-2024	CUST11153	Central	Raipur	Consumer	Online	Furniture	Desks	P:			0.2	1725.85	345.1
ORD114470	08-01-2024	CUST10367	Central	Indore	Consumer	Retail Store	Furniture	Storage	P1039	6	825.6	0.3	4953.6	1486.0
ORD103162	08-01-2024	CUST10729	Central	Nagpur	Consumer	Online	Home & Kitc	Appliances	P1014	7	793.85	0.2	5556.95	1111.3
ORD114099	09-01-2024	CUST10929	Central	Bhopal	Consumer	Retail Store	Office Suppl	Paper	P1091	6	3839.81	0	23038.86	

4.Formatted Dataset



sales_data (1) - Excel

Order_ID	Order_Date	Customer_ID	Region	City	Segment	Channel	Category	Subcategory	Product_ID	Quantity	Unit_Price_INR	Discount	Sales_INR
ORD100000	12-04-2024	CUST11025	Central	Raipur	Small Busin	Retail Store	Technology	Laptops	P1009	6	1144.08	0.15	6864.48
ORD100001	11-03-2025	CUST11950	Central	Nagpur	Corporate	Online	Beauty	Haircare	P1029	7	548.82	0.15	3841.74
ORD100002	27-09-2024	CUST10253	South	Bengaluru	Corporate	Retail Store	Home & Kitc	Cookware	P1043	6	1017.71	0.2	6106.26
ORD100003	16-04-2024	CUST11044	North	Lucknow	Corporate	Online	Office Suppl	Pens	P1037	3	1184.32	0.1	3552.96
ORD100004	12-03-2024	CUST11883	South	Hyderaba	Corporate	Retail Store	Technology	Phones	P1027	6	1593.68	0.05	9562.08
ORD100005	01-12-2025	CUST10230	South	Hyderaba	Consumer	Retail Store	Technology	Laptops	P1018	6	366.69	0.1	2200.14
ORD100006	21-01-2024	CUST11201	East	Bhubanes	Consumer	Online	Beauty	Makeup	P1042	3	345.49	0.05	1036.47
ORD100007	06-09-2025	CUST10655	East	Patna	Consumer	Online	Beauty	Haircare	P1068	1	757.29	0	757.29
ORD100008	01-05-2024	CUST10559	West	Surat	Corporate	Retail Store	Technology	Laptops	P1017	7	642.13	0.25	4494.91
ORD100009	11-04-2025	CUST11190	East	Guwahati	Consumer	Online	Home & Kitc	Decor	P1071	5	445.58	0	2227.9
ORD100010	02-08-2024	CUST10481	North	Jaipur	Consumer	Marketplac	Furniture	Storage	P1011	3	3641.82	0.1	10925.46
ORD100011	26-11-2024	CUST11931	Central	Nagpur	Corporate	Retail Store	Furniture	Storage	P1025	7	2905.71	0	20339.97
ORD100012	03-04-2025	CUST11965	South	Hyderaba	Consumer	Retail Store	Home & Kitc	Cleaning	P1041	4	1289.7	0.05	5158.8
ORD100013	28-03-2024	CUST10944	Central	Indore	Consumer	Retail Store	Office Suppl	Pens	P1017	4	708.41	0.15	2833.64
ORD100014	07-01-2025	CUST10055	North	Delhi	Small Busin	Online	Home & Kitc	Decor	P1038	4	924.67	0	3698.68
ORD100015	09-04-2024	CUST11412	Central	Bhopal	Consumer	Marketplac	Technology	Phones	P1091	1	1146.22	0.05	1146.22
ORD100016	25-10-2025	CUST11905	North	Jaipur	Consumer	Marketplac	Furniture	Tables	P1064	5	446.08	0.05	2230.4
ORD100017	10-05-2024	CUST10414	South	Bengaluru	Consumer	Online	Furniture	Storage	P1007	3	1591.09	0.3	4773.27
ORD100018	23-10-2025	CUST12019	North	Jaipur	Consumer	Retail Store	Furniture	Desks	P1087	7	1750.18	0.1	12251.26
ORD100019	04-11-2024	CUST10195	West	Ahmedab	Consumer	Retail Store	Office Suppl	Binders	P1038	7	1746.07	0.15	12222.49
ORD100020	09-12-2024	CUST12374	Central	Nagpur	Small Busin	Marketplac	Furniture	Chairs	P1029	6	929.3	0.05	5575.8
ORD100021	06-05-2025	CUST10440	South	Kochi	Corporate	Online	Furniture	Chairs	P1028	1	3729.17	0	3729.17
ORD100022	17-02-2025	CUST11923	Central	Indore	Consumer	Online	Furniture	Tables	P1074	1	1310.76	0	1310.76
ORD100023	20-01-2025	CUST11936	East	Kolkata	Consumer	Retail Store	Home & Kitc	Appliances	P1090	1	1465.2	0.05	1465.2
ORD100024	10-07-2024	CUST11803	West	Mumbai	Consumer	Retail Store	Office Suppl	Pens	P1077	6	1932.77	0.1	11596.62
ORD100025	03-10-2024	CUST10456	Central	Bhopal	Consumer	Online	Office Suppl	Pens	P1064	5	1020.13	0	5100.65

6. Sales Data Analysis Using Microsoft Excel

Microsoft Excel was used to perform sales analysis and identify important business trends from the cleaned sales dataset.

Pivot Tables and Pivot Charts were created to summarize sales information and understand business performance from different perspectives.

6.1 Pivot Table Analysis

The following analyses were performed using Pivot Tables:

1. Monthly Sales Analysis – Analyzed sales performance month by month.
2. Year-wise Sales Analysis – Compared sales performance across different years.
3. Region-wise Sales Analysis – Evaluated sales across different geographical regions.
4. Category-wise Sales Analysis – Analyzed sales generated by different product categories.
5. Sub-Category-wise Sales Analysis – Identified product sub-categories contributing to sales.
6. City-wise Sales Analysis – Compared sales performance across different cities.
7. Segment-wise Sales Analysis – Analyzed sales across different customer segments.
8. Channel-wise Sales Analysis – Evaluated sales through different sales channels.
9. Profit Analysis – Examined profit generated from sales transactions.
10. Payment Method Analysis – Analyzed the distribution of sales by payment method.

6.2 Excel Formulas and Calculations

Excel formulas were used to support sales calculations and analysis.

Important calculations included:

- Total Sales
- Total Profit
- Total Quantity
- Discount Amount
- Net Sales
- Profit Margin
- Average Selling Price

6.3 Pivot Chart Creation

Pivot Charts were created to represent the summarized information visually.

The charts helped identify:

- Monthly sales trends.
- Regional sales performance.
- Category-wise sales contribution.
- City-wise sales performance.
- Profitability across products and categories.

6.4 Interactive Slicers

Slicers were added to the Excel dashboard to make data analysis easier.

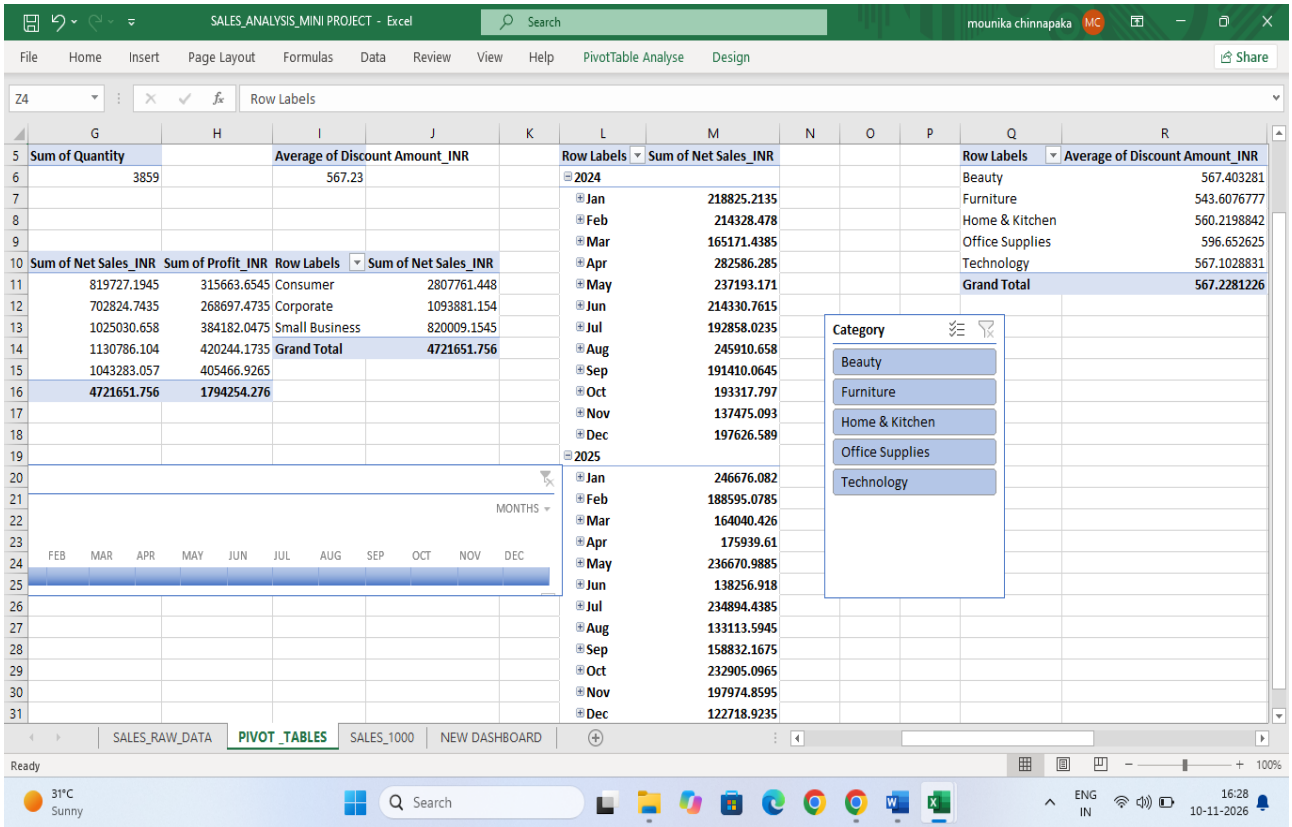
They helped filter sales information based on available fields such as Region, City, Category, and other relevant dimensions.

6.5 Excel Dashboard Development

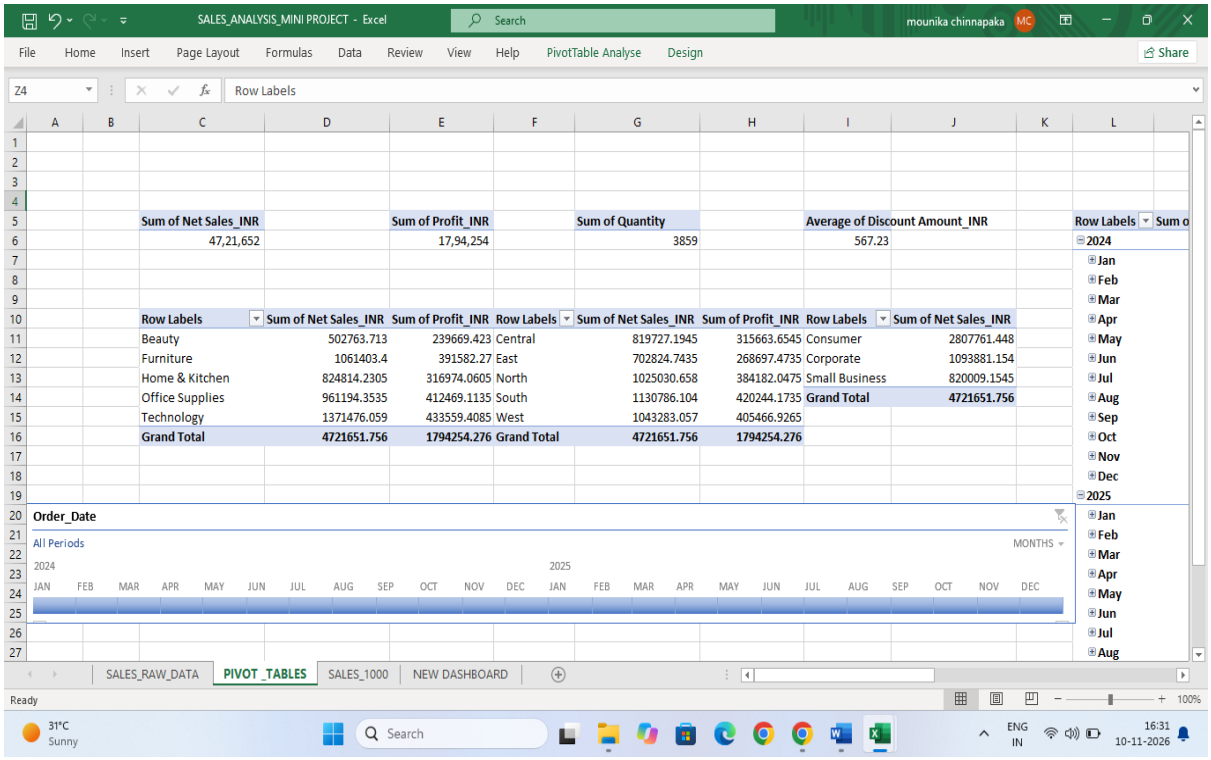
An interactive Sales Analysis Dashboard was created using Pivot Charts, KPI values, and slicers.

The dashboard provides a consolidated view of sales performance and helps users understand important business metrics quickly.

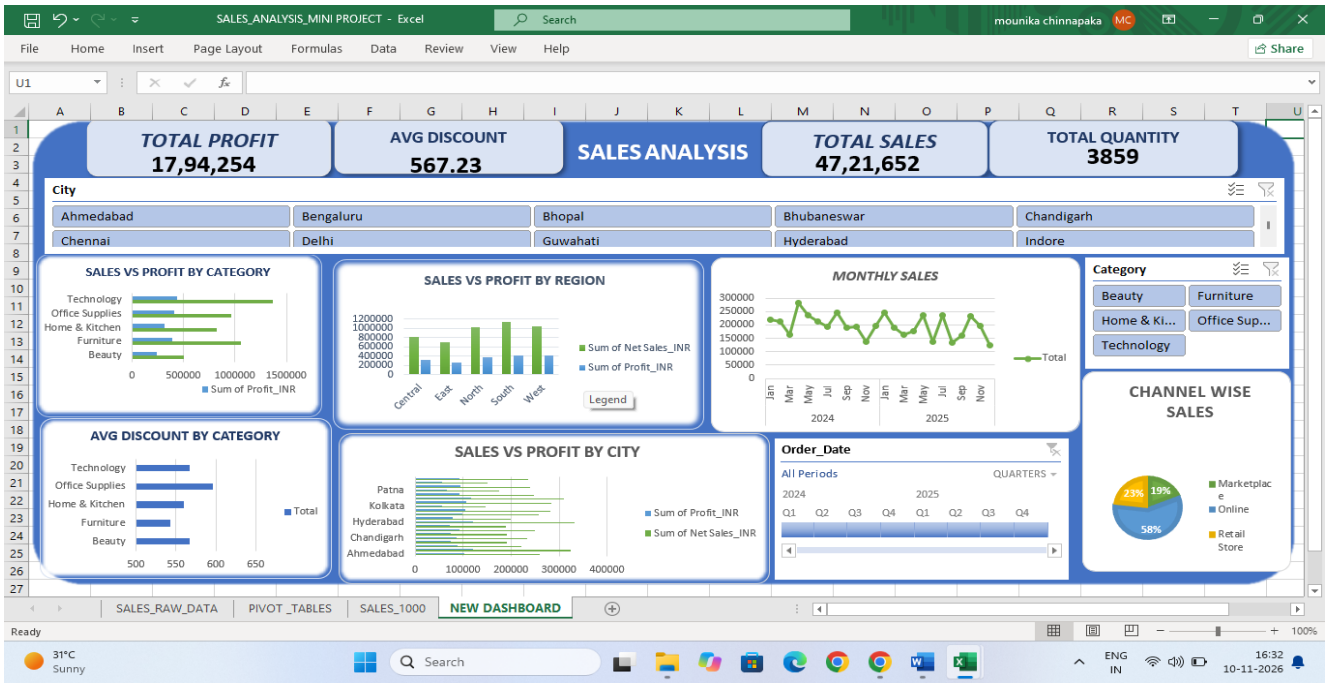
Monthly Sales Pivot Table



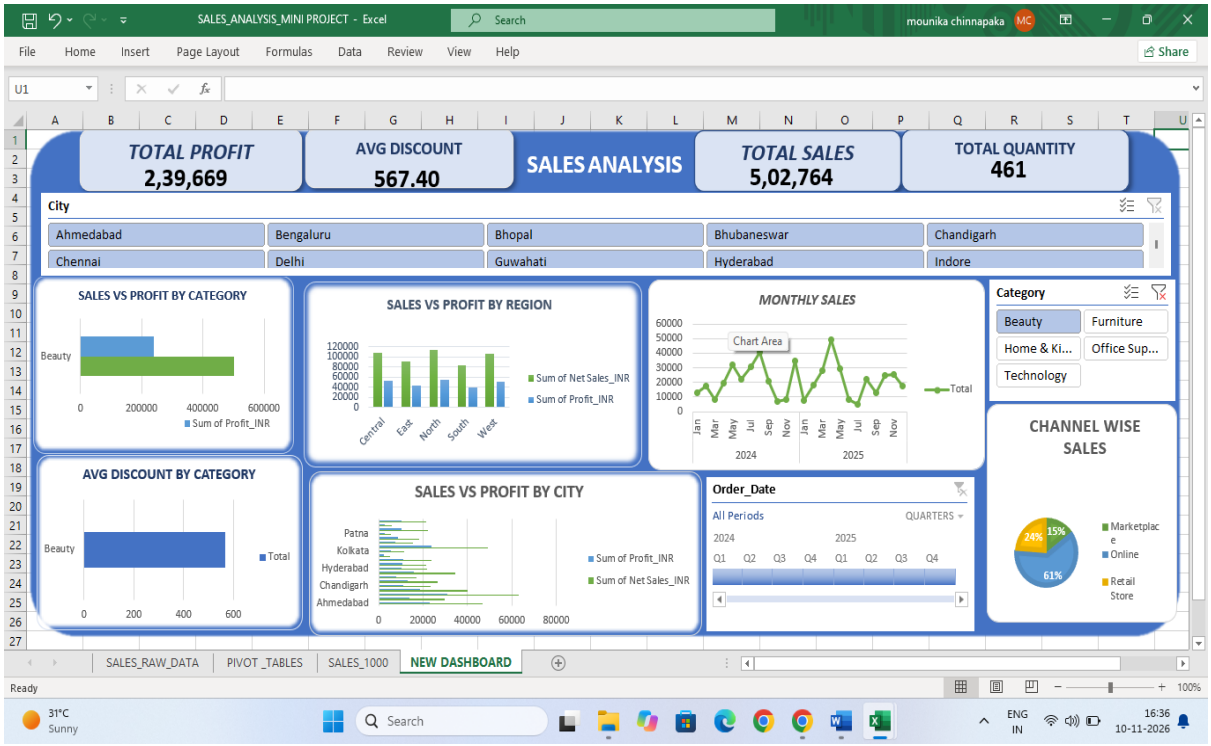
Sales Pivot Table



Dashboard



Interactive Slicers



7. Excel Dashboard Analysis

An interactive Sales Analysis Dashboard was developed using Microsoft Excel to present important business metrics and sales performance in a visual format.

The dashboard was created using Pivot Tables, Pivot Charts, KPI values, and interactive slicers.

The main purpose of the dashboard is to provide a clear overview of business performance and help users analyze sales information efficiently.

7.1 Key Performance Indicators (KPIs)

The dashboard presents important business performance indicators, including:

- Total Sales – Represents the overall sales generated.
- Total Profit – Shows the profit earned from sales transactions.
- Total Quantity – Represents the total quantity of products sold.
- Total Orders – Indicates the number of sales orders.

7.2 Dashboard Visualizations

Different charts were used to represent sales performance:

- Monthly Sales Chart – Shows monthly sales trends.
- Category-wise Sales Chart – Compares sales across product categories.
- Region-wise Sales Chart – Shows sales distribution across regions.
- City-wise Sales Chart – Identifies sales performance by city.
- Profit Analysis Chart – Helps understand business profitability.

7.3 Interactive Dashboard Features

Slicers were incorporated to allow users to filter and explore sales information based on selected categories and geographical dimensions.

The dashboard provides a user-friendly interface for analyzing business performance.

7.4 Benefits of the Excel Dashboard

- Simplifies complex sales information.
- Provides a quick overview of important KPIs.
- Helps identify sales trends and patterns.
- Supports regional and product performance comparisons.
- Improves the presentation of business information.
- Supports data-driven business decisions.

8. Power BI Sales Data Analysis Project

8.1 Project Overview

The Power BI Sales Data Analysis project focuses on analyzing sales performance using Microsoft Power BI. The project transforms raw sales data into meaningful insights through data cleaning, data modelling, DAX calculations, and interactive visualizations.

The dashboard helps users understand sales trends, profitability, regional performance, product performance, and customer-related metrics.

8.2 Tools and Technologies Used

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Excel Dataset
- Data Visualization and Reporting

8.3 Dataset Overview

- Total Records: 1,000
- Total Columns: 24
- File Format: Excel (.xlsx)
- Data Type: Sales Transaction Data

8.4 Data Import and Data Cleaning

Data Import

The sales dataset was imported into Power BI Desktop from an Excel workbook.

Steps performed:

1. Opened Power BI Desktop.
2. Selected Get Data → Excel Workbook.
3. Selected the Sales Analysis dataset.
4. Loaded the dataset into Power Query Editor for data transformation.

Data Cleaning Using Power Query

The following data cleaning operations were performed:

1. Checked column names and data types.
2. Identified and handled missing values.
3. Checked duplicate records.
4. Corrected data type inconsistencies.
5. Verified date and numerical columns.
6. Checked data quality and consistency.
7. Applied the changes using Close & Apply.

Outcome

The raw sales dataset was cleaned and prepared for data modelling and further analysis in Power BI.

8.5 Data Modelling

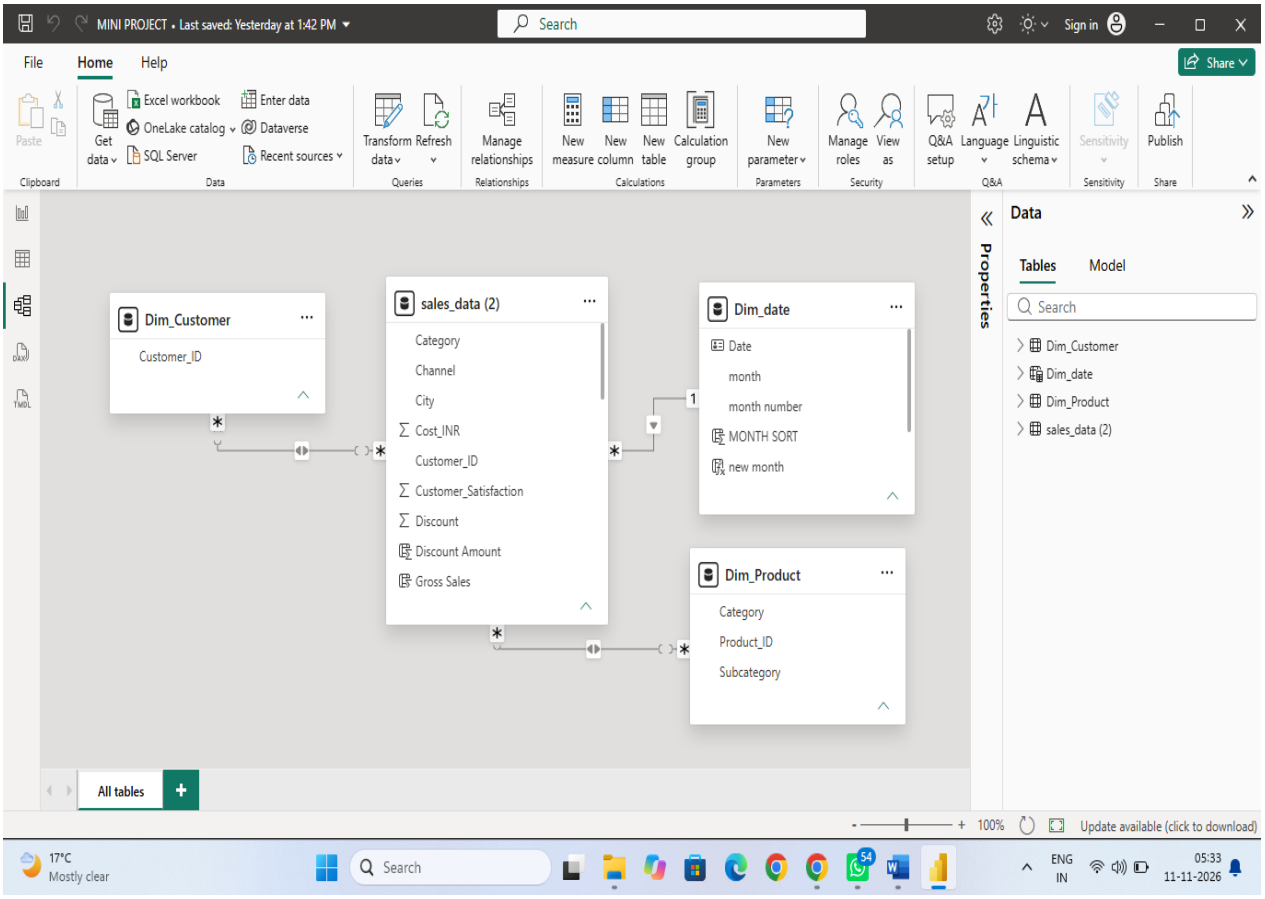
Data modelling is an important step in Power BI that helps establish relationships between tables and organize data for effective analysis.

Activities Performed

1. Opened Model View in Power BI Desktop.
2. Reviewed the available tables.
3. Created and verified relationships between tables.
4. Checked relationship cardinality.
5. Verified filter direction.
6. Ensured the data model supports accurate calculations and visualizations.

Outcome

The data model was prepared to support efficient data analysis, filtering, and reporting in Power BI.

- 
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Search
Sign in
File
Home
Help
Share
Paste
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
New column
New table
Calculation group
New parameter
Manage roles
View as
Q&A setup
Language
Linguistic schema
Sensitivity
Publish
Clipboard
Data
Queries
Relationships
Calculations
Parameters
Security
Q&A
Sensitivity
Share
Dim_Customer
Customer_ID
sales_data (2)
Category
Channel
City
Cost_INR
Customer_ID
Customer_Satisfaction
Discount
Discount Amount
Gross Sales
Dim_date
Date
month
month number
MONTH SORT
new month
Dim_Product
Category
Product_ID
Subcategory
Properties
Data
Tables
Model
Search
Dim_Customer
Dim_date
Dim_Product
sales_data (2)
All tables
100%
Update available (click to download)
17°C
Mostly clear
Search
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MODEL VIEW

8.6 DAX Measures

DAX (Data Analysis Expressions) is a formula language used in Power BI to perform calculations and create meaningful business metrics.

DAX Measures Created

A total of 10 DAX measures were created to analyze sales performance and profitability.

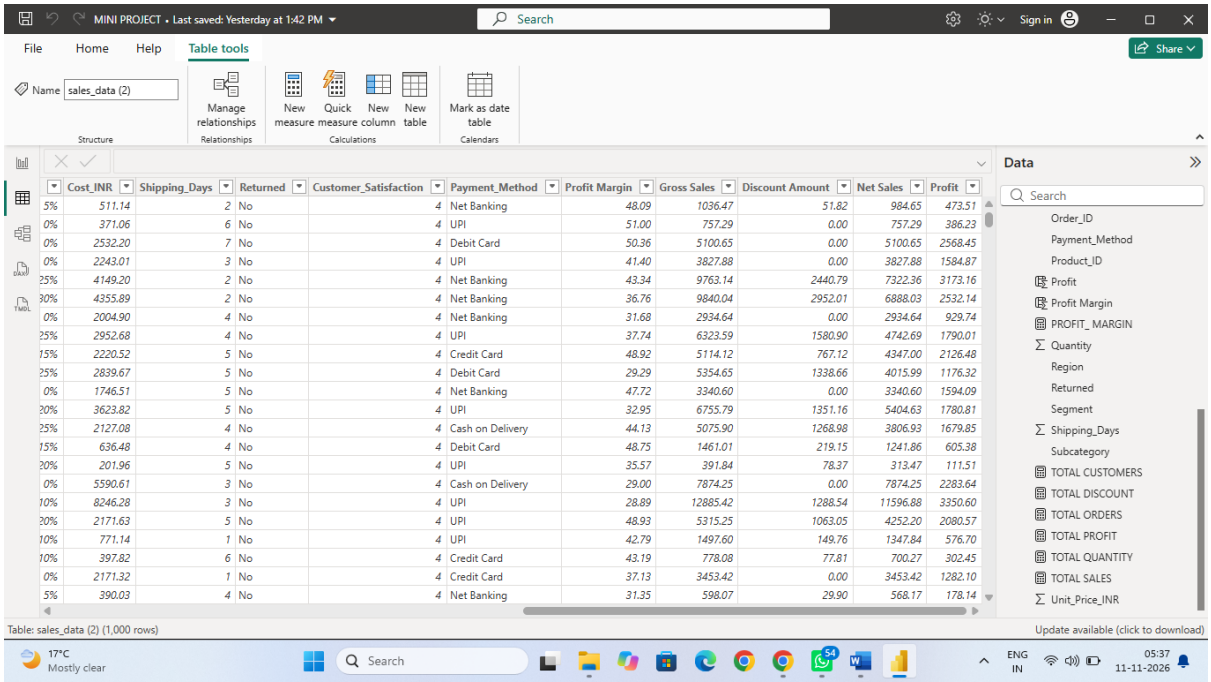
These measures help calculate key business metrics such as:

- Total Sales
- Total Profit
- Total Quantity
- Total Orders
- Total Customers
- Average Sales
- Total Discount
- Profit Margin
- Average Selling Price
- Other sales performance indicators

Purpose of DAX Measures

- To calculate important business KPIs.
- To analyze sales and profit performance.
- To support interactive dashboard filtering.
- To provide meaningful insights for business decision-making.

Outcome:DAX measures were successfully created and used in KPI cards and Power BI visualization



Cost_INR	Shipping_Days	Returned	Customer_Satisfaction	Payment_Method	Profit Margin	Gross Sales	Discount Amount	Net Sales	Profit
5%	511.14	2	No	4	Net Banking	48.09	1036.47	51.82	984.65	473.51
0%	371.06	6	No	4	UPI	51.00	757.29	0.00	757.29	386.23
0%	2532.20	7	No	4	Debit Card	50.36	5100.65	0.00	5100.65	2568.45
0%	2243.01	3	No	4	UPI	41.40	3827.88	0.00	3827.88	1584.87
25%	4149.20	2	No	4	Net Banking	43.34	9763.14	2440.79	7322.36	3173.16
30%	4355.89	2	No	4	Net Banking	36.76	9840.04	2952.01	6888.03	2532.14
0%	2004.90	4	No	4	Net Banking	31.68	2934.64	0.00	2934.64	929.74
25%	2952.68	4	No	4	UPI	37.74	6323.59	1580.90	4742.69	1790.01
15%	2220.52	5	No	4	Credit Card	48.92	5114.12	767.12	4347.00	2126.48
25%	2839.67	5	No	4	Debit Card	29.29	5354.65	1338.66	4015.99	1176.32
0%	1746.51	5	No	4	Net Banking	47.72	3340.60	0.00	3340.60	1594.09
20%	3623.82	5	No	4	UPI	32.95	6755.79	1351.16	5404.63	1780.81
25%	2127.08	4	No	4	Cash on Delivery	44.13	5075.90	1268.98	3806.93	1679.85
15%	636.48	4	No	4	Debit Card	48.75	1461.01	219.15	1241.86	605.38
20%	201.96	5	No	4	UPI	35.57	391.84	78.37	313.47	111.51
0%	5590.61	3	No	4	Cash on Delivery	29.00	7874.25	0.00	7874.25	2283.64
10%	8246.28	3	No	4	UPI	28.89	12885.42	1288.54	11596.88	3350.60
20%	2171.63	5	No	4	UPI	48.93	5315.25	1063.05	4252.20	2080.57
10%	771.14	1	No	4	UPI	42.79	1497.60	149.76	1347.84	576.70
10%	397.82	6	No	4	Credit Card	43.19	778.08	77.81	700.27	302.45
0%	2171.32	1	No	4	Credit Card	37.13	3453.42	0.00	3453.42	1282.10
5%	390.03	4	No	4	Net Banking	31.35	598.07	29.90	568.17	178.14

8.7 KPI Cards and Data Visualizations

KPI Cards

Four KPI cards were created in Power BI to display important business performance indicators.

KPI cards provide a quick overview of key sales and profitability metrics.

Data Visualizations Created

The following charts were developed in the Power BI Sales Dashboard:

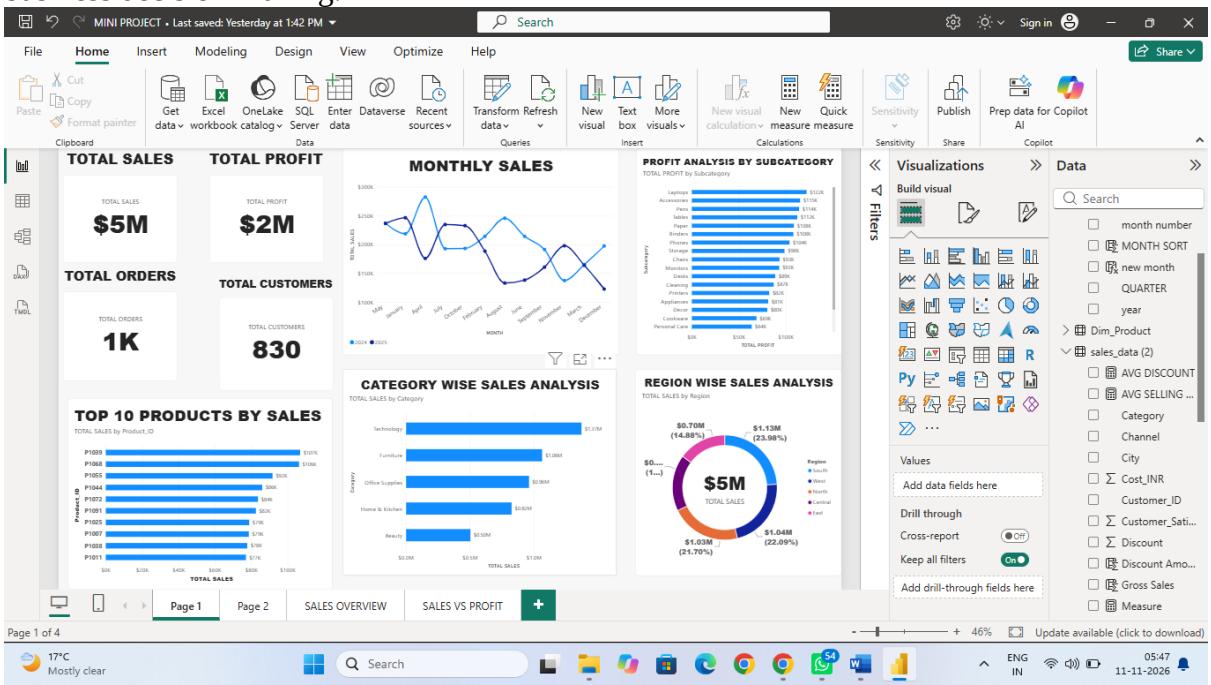
1. Category-wise Sales – Bar Chart
2. Region-wise Sales – Donut Chart
3. Profit Analysis by Sub-Category
4. Top 10 Cities by Sales
5. Discount vs Profit – Scatter Chart

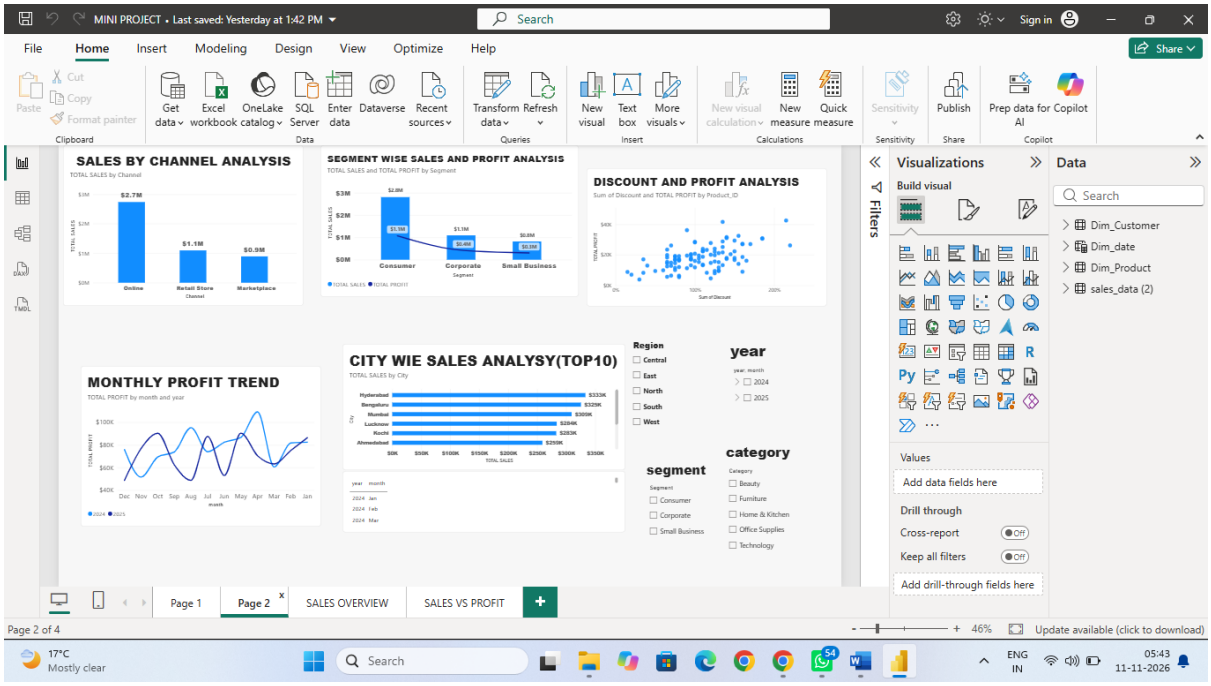
Purpose of Visualizations

- To compare sales across product categories.
- To understand regional sales distribution.
- To identify profitable and less profitable sub-categories.
- To identify the top-performing cities.
- To analyze the relationship between discount and profit.

Outcome

Interactive visualizations were created to make sales analysis easier and support better business decision-making.




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DISCOUNT AND PROFIT ANALYSIS
MONTHLY PROFIT TREND
CITY WIE SALES ANALYSY(TOP10)
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Dim_date
Dim_Product
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Values
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Drill through
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8.8 Slicers and Interactive Dashboard

Interactive Slicers

Slicers were added to the Power BI dashboard to provide interactive filtering capabilities.

Users can filter the sales data based on selected categories and analyze specific business performance.

Slicer Functionality

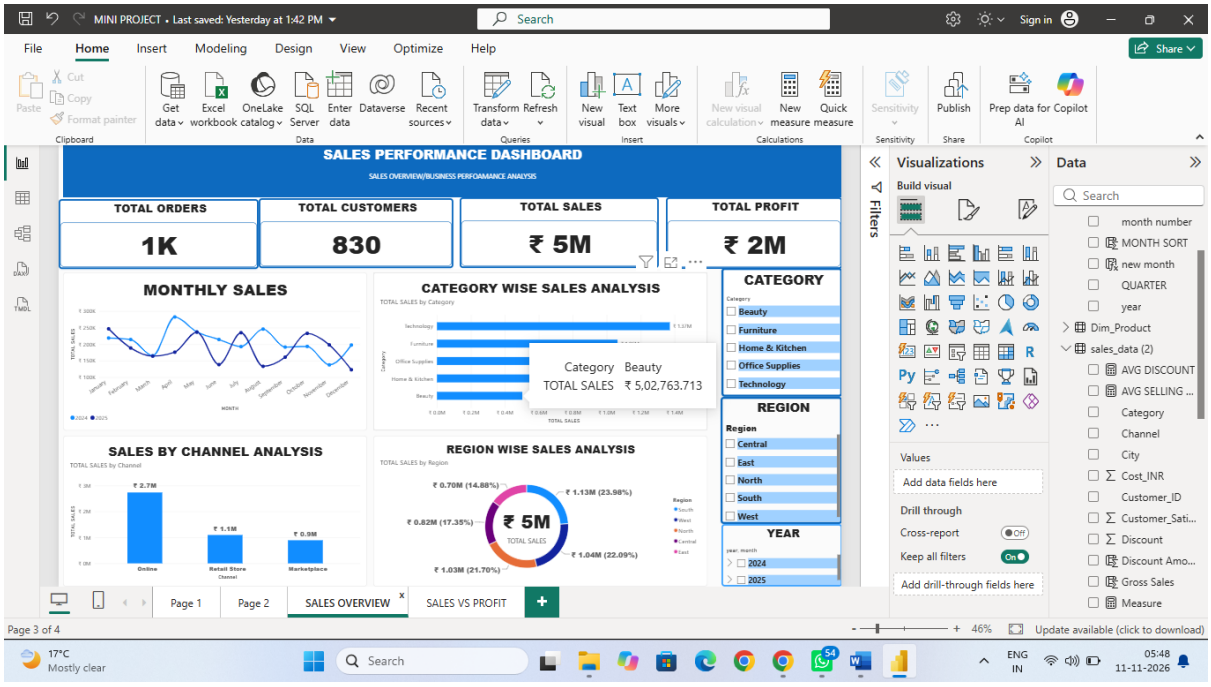
- Slicers allow users to filter dashboard data dynamically.
- Multiple charts respond to the selected filter values.
- Report interactions help users explore sales performance from different perspectives.
- Sync Slicers functionality can be used to maintain consistent filters across multiple report pages.

Dashboard Interactivity

The Power BI dashboard provides an interactive environment where users can explore sales, profit, categories, regions, and city-level performance.

Outcome

Interactive filtering improves the usability of the dashboard and helps users perform detailed sales analysis.



8.9 Business Insights and Findings

The Power BI Sales Analysis dashboard helps identify important business trends and performance indicators.

Key Business Insights

1. **Sales Performance:** Analyzes overall sales performance and identifies sales trends.
2. **Category Analysis:** Helps compare sales across different product categories.
3. **Regional Performance:** Identifies regions contributing to sales.
4. **Profitability Analysis:** Helps identify profitable and less profitable product sub-categories.
5. **City Performance:** Highlights the top 10 cities based on sales.
6. **Discount Analysis:** Helps understand the relationship between discounts and profitability.
7. **Interactive Analysis:** Enables users to explore business performance using slicers.

8.10 Project Conclusion

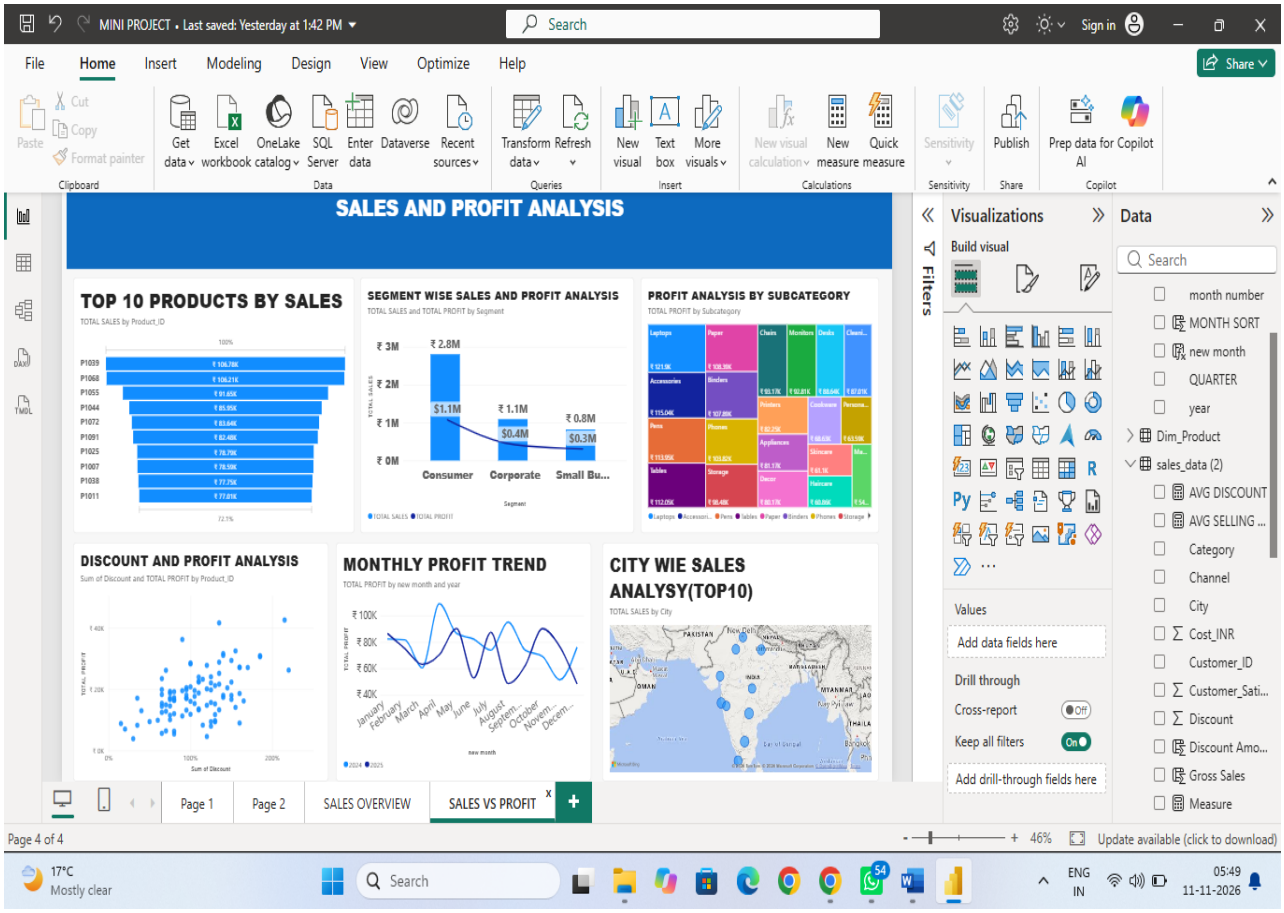
The Sales Data Analysis project was successfully completed using Microsoft Excel and Power BI.

Excel was used for data cleaning, analysis, Pivot Tables, Pivot Charts, and interactive dashboard creation.

Power BI was used for data transformation, data modelling, DAX calculations, KPI cards, and interactive visualizations.

This project improved practical knowledge of Excel, Power Query, Power BI, DAX, data visualization, and business analysis.

The project demonstrates how raw sales data can be transformed into meaningful business insights to support data-driven decision-making.


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DASHBOARD